Geometric Configuration and Dynamical Degrees of Freedom of the Vacuum

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This paper presents the derivation process and logic of the author's two-layer, topologically adjacency-invariant, fixed-chirality background elastic space model since February 2025:

- Space is the ontological substrate in which physical processes occur and evolve; gravitational geometry is the evolution of space. The minimal quantization of the other three fundamental interactions occurs at the Planck scale and is also the evolution of the space ontology itself.
- From the constancy of the speed of light in vacuum and the smooth curvature of light rays around black holes, it follows that space should be a geometry with invariant topological adjacency relations. If the adjacency relations of the space quantum network were dynamically changing, it would be difficult to explain why the speed of light in vacuum is constant.
- However, no Planck-scale quantization signatures have been observed in gravitational waves; moreover, the quantum spin at the Planck scale conflicts with the metric geometric degrees of freedom of general relativity; furthermore, there is no phenomenon supporting the screening of gravity; and additionally, to explain the energy conservation problem of gravitational redshift, it follows that the substrate mediating gravity is at a deeper level — leading to the derivation of a sub-Planck metric layer.
- The quantized network at the Planck layer must have at least resonance and spin as degrees of freedom, because these are the degrees of freedom required for electromagnetic interactions, and electromagnetic waves propagate directly at the Planck scale, which is the minimal scale at which quantization phenomena occur. The axis-rotation degree of freedom can be derived simply: the speed of light in vacuum is isotropic, and if the spin axis were fixed, it would be impossible to cover the symmetry of space.
- From the phenomenon of parity non-conservation, it follows that space should possess some kind of fixed-chirality background. In this model, this is the fixed-chirality

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ground-state quantum spin. This postulate can also cover the 3D intuitive picture of matter-antimatter asymmetry and the Higgs mechanism. Therefore, this postulate of a fixed-chirality quantum spin is economical and consistent with phenomena.

- This paper only discusses the invariance of adjacency relations, without going into the specific adjacency configuration. To ensure isotropy at the observational level, it is possible to achieve isotropy at observational scales through the statistical averaging of a randomly dense distribution.
- The above discussion shows that it is not necessary to introduce extra dimensions, and that the four fundamental interactions can nevertheless be covered. Detailed content is as follows, or refer to the 2025 preprint.

Abstract

Based on the chapter on space configuration and elementary quantum degrees of freedom in Zou, Zhi Kai's 2025 preprint [1], this article systematically elaborates the setting logic of spacetime configuration and dynamical degrees of freedom in this model. The model takes a topologically invariant 3D Space Elementary Quanta (SEQ) network as its ontological substrate, assigning each SEQ three intrinsic degrees of freedom—resonance frequency, resonance axis vector, and spin—and assigning adjacent SEQs interactive variables such as spacing, tensor deformation, twistor deformation, and deformation energy. This article argues that the design purpose of this seemingly simple set of 3D dynamical degrees of freedom is precisely to cover the metric degrees of freedom of traditional gravitational theory (GR) and the gauge degrees of freedom of quantum field theory (QFT) **without introducing extra dimensions and without breaking 3D topological invariance**, thereby attempting to uniformly describe all fundamental interactions with a single 3D network dynamics. The article will analyze one by one the physical connotations of spin, axis rotation, resonance, tensor deformation, and twistor deformation, explaining how they respectively correspond to SU(3) color compression, SU(2) chiral rotation, U(1) electromagnetic vorticity, the Higgs locking mechanism, and gravitational elastic response, and will discuss the open problems this setting faces in mathematical formalization.

Keywords: SEQ network; dynamical degrees of freedom; spin; axis rotation; resonance; tensor deformation; twistor deformation; gauge symmetry; quantum

gravity

I. Introduction: Why We Need to Reset the Dynamical Degrees of Freedom of Spacetime

The core dilemma facing contemporary physics lies in the ontological incompatibility between general relativity and quantum field theory. GR treats spacetime as a continuous, differentiable, dynamically curved Lorentzian manifold, whose fundamental dynamical degree of freedom is the metric tensor g_μν; QFT treats matter and interactions as quantum fields defined on a fixed spacetime background, whose fundamental dynamical degrees of freedom are gauge connections and field operators on fiber bundles. Each has achieved brilliant success, yet they are difficult to coexist within the same ontological framework.

Zou Zhi Kai's MITC model proposes a unique solution path: **not introducing extra dimensions, not abandoning the ontological reality of 3D space, but rather resetting the fundamental constituent units of 3D space itself and their dynamical degrees of freedom**, so that the geometric degrees of freedom of GR and the gauge degrees of freedom of QFT both become different manifestations of the same 3D network dynamics.

The core output of this setting is: it refuses to treat "extra dimensions" as physical reality, but instead **reinterprets the dynamical degrees of freedom encoded in extra dimensions in higher-dimensional theories as the internal degrees of freedom of 3D spatial lattice points (SEQ) themselves**.

This "dimensionality-reduction reconstruction" strategy enables the model to accommodate all the gauge structures required by the Standard Model while maintaining 3D topological invariance. The purpose of this article is to systematically sort out the physical connotations and unified logic of each dynamical degree of freedom in this setting.

II. Ontological Substrate: The Topologically Invariant SEQ Network

Before unfolding the discussion of dynamical degrees of freedom, the substrate setting of the model must first be clarified.

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2.1 Space Elementary Quanta (SEQ) and the Dual-Layer Substrate

The model assumes that cosmic space is a 3D network composed of discrete "Space Elementary Quanta" (SEQ). Each SEQ is a Planck-scale, topologically stable structural unit, composed of deeper sub-Planckian elastic constituents. These SEQs are interconnected via spring-like bonds, forming a **topologically homeomorphic 3D network**: adjacency relations never change, no node gains or loses neighbors, and no connection is ever broken or newly created.

This setting contains two layers:

- **Sub-Planckian layer**: the elastic substrate constituting SEQ, responsible for the elastic response of space (storage and release of elastic potential energy);

- **Planckian layer (SEQ layer)**: the topologically stable structure composed of sub-Planckian constituents, responsible for quantized energy transfer.

The key to this dual-layer structure is: **the spin degrees of freedom and the elastic degrees of freedom decouple in low-energy states**. The spin of SEQ is a topologically protected excitation on the sub-Planckian topological network, and its dynamics do not interfere with the elastic deformation between SEQs; conversely, elastic deformation does not destroy the topological protection of spin. This decoupling mechanism allows quantum spin and spatial elasticity to coexist in the same 3D network without contradiction. It can also explain where the energy lost by the lowered electromagnetic wave frequency in gravitational redshift goes, ensuring energy conservation.

2.2 Global Topological Invariance as the Core Postulate

The core postulate of the model is: **the topologically homeomorphic structure of space remains invariant during physical evolution**. The reasonableness of this postulate comes from two aspects:

- **Phenomenological basis**: the light bending around black holes exhibits a highly smooth distribution of spacetime curvature, with no observed topological tearing or singular connectivity changes; if the spatial topological adjacency relations were variable, it would be very difficult to be consistent with the phenomenon of constant light speed in vacuum.

- **Theoretical basis**: the Feynman path integral requires weighted integration over all possible paths, which presupposes that a well-defined measure exists on the path space. If the spatial topology changed dynamically, the set of paths would no longer have a consistent measure, and the integral would lose meaning.

This topological invariance postulate provides the model with three key corollaries: an intuitive physical picture of the constancy of the speed of light, a concrete mechanism for global energy conservation, and a great simplification of the mathematical form of the Hamiltonian.

III. The Dynamical Degrees of Freedom of SEQ

On the topologically invariant SEQ network, each SEQ is assigned three intrinsic dynamical degrees of freedom:

3.1 Resonance Frequency ω_i

Each SEQ is in a ground-state resonance state, and its resonance frequency ω_i is the first generalized coordinate. In the ground state (without any deformation), the resonance frequency of all SEQs equals the Planck frequency ω , corresponding to the reciprocal of the Planck time t , i.e., the maximum transformation frequency of the universe in theory.

The physical meaning of the resonance frequency lies in: **it directly encodes the geometric deformation information of local space**. Any form of metric change—compression, stretching, twisting—will modulate the resonance frequency of SEQ, causing it to deviate from the Planck frequency. This "geometry-frequency mirroring" principle is the core bridge connecting GR and QFT in the model.

The change in resonance frequency follows nonlinear, asymmetric elastic response laws: the compression phase and the stretching phase have different elastic coefficients, so their modulation strengths on frequency differ. This asymmetry provides a concrete mechanism for understanding phenomena such as gravitational redshift and time dilation.

3.2 Axis Rotation: Resonance Axis Vector (i)

The resonance of each SEQ is not isotropic, but along a specific direction—the resonance axis vector (i). This is the second generalized coordinate.

The physical meaning of the resonance axis vector lies in: **it encodes the anisotropy and directional structure of local space**. In the ground state, the direction of the resonance axis is determined by the adjacency configuration of SEQ, and is therefore discrete and finite, which means that the resonance axis direction itself is also quantized.

The resonance axis vector and the resonance frequency together constitute the two generalized coordinates (ω_i, (i)) in the Hamiltonian formalism. The model assumes that the distribution of the spatial potential energy $U(\omega_i$, (i)) may determine the orientation of the resonance axis in space. This assumption links the potential energy term with the axial structure, providing a clue for understanding the geometric origin of gauge fields.

3. 3 Spin: Topologically Protected Torsion Mode on the Sub-Planckian Substrate

Each SEQ also carries a spin degree of freedom. In the model, spin is explicitly defined as a **topologically protected torsion mode on the sub-Planckian substrate**. This means:

- Spin is not an independent property attached to SEQ, but a collective excitation on the sub-Planckian topological network;
- Spin and resonance oscillation mutually excite each other and maintain phase synchronization;
- Spin is always orthogonal to the resonance axis—this orthogonality is the geometric origin of the transverse wave nature of electromagnetic waves and the fixed chirality of the SEQ background.

Most critically, the model assumes that **the ground-state spin chirality of SEQ is**

fixed. This assumption directly explains parity non-conservation: if the fundamental constituent units of space have a definite chirality from the very bottom, then left-right symmetry is broken from the start. At the same time, fixed chirality also provides a physical picture for matter-antimatter asymmetry: the structural matrix of antimatter couples differently with the fixed-chirality ground state of SEQ, causing its structural matrix to be unstable.

IV. Interactive Variables Between Adjacent SEQs

In addition to the intrinsic degrees of freedom of each SEQ, the model also defines the following interactive variables for the interactions between adjacent SEQs:

4.1 Spacing

Spacing is the equilibrium distance between adjacent SEQs. In the ground state, adjacent SEQs maintain a dynamic equilibrium spacing through spring-like bonds. When space is compressed, the spacing decreases; when space is stretched, the spacing increases.

The physical meaning of spacing change lies in: **it directly corresponds to gravitational interaction**. The model models gravity as a "translational action"—changing the equilibrium spacing between SEQs. When a mass source compresses local space, the spacing of the surrounding space is stretched, generating tension and finite potential energy. This system is like a loaded spring: under a gravitational field, the oscillation frequency decreases, the local spacetime transformation rate decreases, causing time dilation—this is consistent with the predictions of general relativity, while revealing its microscopic mechanism.

4.2 Tensor Deformation

Tensor deformation encodes the **directional elastic stress** between adjacent SEQs. When space is subjected to different compression or stretching in different directions, tensor deformation describes this anisotropic deformation pattern.

The physical meaning of tensor deformation lies in: **it corresponds to the degrees of freedom of the metric tensor in GR**. In the continuous limit, the synthetic effect of

tensor deformation can reproduce the changes of the metric tensor. The model explicitly states:

"The dynamic deformation of 3D space structural matrix and variation in local SEQ density distribution corresponds to Metric field in General Relativity."

This means that the metric degrees of freedom of GR are not an additional structure independent of the SEQ network, but rather an effective description of the tensor deformation between SEQs in the macroscopic continuous limit.

4.3 Twistor Deformation

Twistor deformation encodes the **torsional coupling between spin and geometry** between adjacent SEQs. When a relative torsion occurs between the spin of SEQ and the resonance axis, twistor deformation describes this coupling mode.

The physical meaning of twistor deformation lies in: **it corresponds to the degrees of freedom of spin connection and gauge fields in QFT**. In Einstein-Cartan theory, spin directly defines the torsion part of the connection; in the model, twistor deformation is precisely the discrete counterpart of this torsion connection. More importantly, twistor deformation is closely related to the Higgs mechanism: the model understands the Higgs mechanism as a "chiral quantum lock"—the coupling of twistor deformation with the fixed chiral spin of SEQ locks the compression state of local space, preventing energy dissipation.

The model further points out that the combined effect of the Kerr metric and the spin connection in Einstein-Cartan theory actually corresponds to the essence of the Higgs mechanism. This provides a geometric intuition for understanding the Higgs mechanism: it is like a screw cap with fixed chirality, where only rotation in a specific direction can tighten and form a stable configuration.

4.4 Deformation Energy

Both tensor deformation and twistor deformation have corresponding deformation energies, which can explain the energy conservation problem in gravitational redshift.

V. The Unified Coverage Logic of Dynamical Degrees of Freedom

Now we can clearly see the design logic of the model: **using a unified set of 3D network dynamical degrees of freedom to simultaneously cover the geometric degrees of freedom of GR and the gauge degrees of freedom of QFT**.

5.1 Covering GR: Metric Degrees of Freedom → Tensor and Twistor Deformation + Frequency Modulation

The core assertion of the model is: **all geometric-dynamical effects of the metric in GR can be fully represented by the frequency field ω_i of the SEQ network and its gradient, as well as the tensor-twistor deformation field**. Specifically:

"The temporal geometry in the 4D formalism of general relativity corresponds precisely to the frequency gradient distribution of the spatial sites in 3D space."

This means that the 4D metric geometry of GR is reconstructed in the model as the frequency gradient distribution of the 3D SEQ network. Gravitational redshift, time dilation, light bending, Mercury's perihelion precession—all these GR effects become macroscopic manifestations of SEQ frequency modulation and tensor deformation.

5.2 Covering QFT: Gauge Degrees of Freedom → Spin, Resonance, Axis Rotation + Deformation + Transverse Wave of Deformation Conduction

The core structure of QFT is gauge symmetry, and the gauge group of the Standard Model is SU(3) × SU(2) × U(1). In the model, these abstract gauge groups are given concrete 3D geometric meanings:

5.2.1 SU(3): Local Space Compression and 3D Modulation

The model reinterprets SU(3) color interaction as **the dynamics of local space compression**. Specifically:

- The 8 generators of SU(3) correspond to 8 different gluon dynamical modes;
- Among them, the 6 off-diagonal matrices represent combinations of color

exchange operations and compression/stretching phase transitions;

- The 2 diagonal matrices correspond to scaling transformations across the three color dimensions;

- These gluon action representations and their interaction types operate in the interlayer regions of the multi-layered structural matrices of protons/neutrons.

Critically, the compression mediated by SU(3) performs 3D modulation and axis transformation while maintaining quasi-spherical symmetry, so the generated gravitational field is also spherically symmetric and divergent. This provides a geometric foundation for the spherically symmetric gravitational effect of mass.

5.2.2 SU(2): Rotation Axis Adjustment and Chirality Encoding

The model reinterprets SU(2) weak interaction as **the rotation axis adjustment of the structural matrix of charged microscopic particles**. The physical content of SU(2) lies in:

- Adjusting the rotation axis and spin of charged particles;

- Encoding the rotational chirality of the structural matrix;

- Rotational structural matrices with different chirality have different coupling modes with the fixed chiral ground-state spin of SEQ.

This provides a geometric picture for parity violation in weak interactions: chirality is not an abstract internal quantum number, but a geometric property of the rotation mode of the structural matrix.

5.2.3 U(1): Electromagnetic Vorticity and Resonance Coupling

The model reinterprets U(1) electromagnetic interaction as **the synchronous mutual excitation of SEQ resonance and spin**. Electromagnetic waves are understood as:

- SEQ resonance generates an electric field;

- The orthogonal torsion of SEQ spin generates a magnetic field;

- The two mutually excite each other and propagate in the form of a transverse wave at the speed of light in the elastic network.

This picture returns to the intuition of Maxwell's original vortex model, while endowing the electromagnetic field with a clear geometric ontological status.

5.2.4 Higgs Mechanism: Chiral Quantum Lock

The Higgs mechanism is reinterpreted in the model as a **chiral quantum lock**—a combination of preloaded torsional spring energy storage and a ratchet mechanism. Its physical content is:

- The compression mediated by SU(3) is like a tensioned spring, storing elastic potential energy;

- The Higgs field is like a ratchet mechanism, locking the compression state through chiral coupling, preventing energy dissipation;

- This locking mechanism stabilizes the mass structure, while making the structural matrix of antimatter unstable.

The model further points out that the Higgs mechanism is essentially a synergistic effect of gauge fields (U(1), SU(2), SU(3)), rather than an independent field. The torsional-spring-like vorticity comes from the vorticity coupling of gauge fields, and the ratchet-like locking comes from the symmetry breaking induced by the fixed chiral spin of SEQ.

All interactions—gravity, strong, weak, electromagnetic—are unified as **different elastic response modes of the same 3D SEQ network**. Gravity is space stretching, the strong force is space compression, the weak force is axis rotation adjustment, and the electromagnetic force is resonance-spin coupling. Mass is the locking of the compression state, Higgs is the locking mechanism, dark matter is a weakly compressed locked state, and dark energy is the ground-state tension of the network.

VI. Preliminary Framework for Mathematical Formalization

The model provides a preliminary Hamiltonian form for this setting of dynamical degrees of freedom:

$$H(\omega_i, \quad (i)) = \Sigma_i K(\omega_i, \quad (i)) + \Sigma_i U(\omega_i, \quad (i))$$

Where:

- $K(\omega_i, \quad (i))$: the energy transport capacity determined by the frequency gradient and directionality (kinetic-like);

- $U(\omega_i, \quad (i))$: the energy storage arising from spatial elastic deformation or gauge coupling (potential-like).

The physical picture of this Hamiltonian is extremely clear: **it is an instantaneous panoramic snapshot of the spatial energy distribution**. Each SEQ contributes its resonance frequency and axis direction, and energy can only flow from high-energy nodes to low-energy adjacent nodes, so this snapshot completely determines all possible system updates at the next moment.

The model further proposes that the spatial distribution of the potential energy term U may determine the orientation of the resonance axis in space. This assumption links potential energy with the axial structure, providing a clue for the geometric origin of gauge fields.

On the basis of the generalized coordinates $(\omega_i, \quad (i))$, the model can construct physical quantities such as momentum, velocity, acceleration, force, kinetic energy, and potential energy, all expressed as polynomials of the generalized coordinates or their differentials. This provides a framework for constructing conservation equations. For details, see ZOU (2025) preprint.

VII. Conclusion

Zou Zhi Kai's MITC model proposes an original unification scheme: **taking a topologically invariant 3D SEQ network as the ontological substrate, by assigning each SEQ three intrinsic degrees of freedom—resonance frequency,**

resonance axis vector, and spin—and assigning adjacent SEQs three interactive variables—spacing, tensor deformation, and twistor deformation—it attempts to cover the geometric degrees of freedom of GR and the gauge degrees of freedom of QFT with a single 3D network dynamics.

The core output of this setting is: **extra dimensions are not physical reality, but mathematical encoding of the internal degrees of freedom of 3D spatial lattice points.** SU(3) is not an abstract symmetry group, but the 3D modulation of local space compression; SU(2) is not abstract weak isospin, but rotation axis adjustment and chirality encoding; U(1) is not abstract phase symmetry, but synchronous mutual excitation of resonance and spin; Higgs is not an independent field, but the chiral quantum lock of gauge field synergy; gravity is not an independent force, but the elastic response of space.

The goal of this framework is: **to unify all interactions as different elastic response modes of the same 3D spatial network.** If this program can complete rigorous mathematical formalization and obtain experimental verification, it will provide a completely new solution path for the quantum gravity problem—not by introducing extra dimensions or non-locality, but by resetting the dynamical degrees of freedom of 3D space itself.

As the model quotes Lao Tzu: "The Way of Heaven diminishes the surplus to supplement the deficient." The dynamics of the universe may precisely be the irreversible redistribution of energy from high to low, from surplus to deficient, on a topologically invariant network. And spin, axis rotation, resonance, tensor deformation, and twistor deformation are precisely the complete expression of degrees of freedom of this redistribution process in 3D space.

Author's Previous Related Works:

1. Previous Preprint from 2025 :Zou, Z. K. (2025). The Thermodynamic Nature of Time, The Geometric Essence of Gravity-Mass, The Quantum Chirality of Space, The Non-Statistical Formula of Entropy, The Dynamical Rules of Causality. Zenodo. https://doi.org/10.5281/zenodo.14788393
2. Zou, Z. K. (2026). The Longitudinal Wave Problem in the Elastic Space Paradigm. Zenodo. https://doi.org/10.5281/zenodo.22741659

3. Latest Update: Zou, Z. K. (2026). Energy-Gradient-Driven Computational Universe: Time-Entropy Mapping; Mass-Gravity Duality; Metric-Frequency Mirroring V39. Zenodo. https://doi.org/10.5281/zenodo.21991046
4. Zou, Z. K. (2026). Demystifying Extra Dimensions via 3D Lattice Dynamics. Zenodo. https://doi.org/10.5281/zenodo.20576855
5. Zou, Z. K. (2025). A prediction of a difference in the magnetic moments of the positron and electron, and A speculative understanding of the mechanisms behind CP violation. Zenodo. https://doi.org/10.5281/zenodo.15080971